AMC Grade 7-8 Preparation For Round 2

1. Which of the following is the greatest value?

A) $\sqrt{\sqrt[3]{5 \cdot 6}}$ B) $\sqrt{6\sqrt[3]{5}}$ C) $\sqrt{5\sqrt[3]{6}}$ D) $\sqrt[3]{6\sqrt{5}}$

$\sqrt[6]{30}$ $\sqrt{\sqrt[3]{6^3 \times 5}}$ $\sqrt{\sqrt[3]{5^2 \times 6}}$ $\sqrt[3]{\sqrt{6^2 \times 5}}$

$\sqrt[6]{1080}$ $\sqrt[6]{750}$ $\sqrt[6]{180}$

$\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$

2. Four numbers, when added to three numbers combined at the same time result in 180, 197, 208, and 222. What is the largest four numbers?

A) 66 B) 96 C) 89 D) 52

Let the 4 numbers be a, b, c, d

When we add 3 of them together at a time we get 180, 197, 208, 222

$$\begin{cases} a+b+c = 180 \\ a+b+d = 197 \\ a+c+d = 208 \\ b+c+d = 222 \end{cases}$$

$+$

$3a+3b+3c+3d = 807$

$a+b+c+d = \frac{807}{3} = 269$

- $a = (a+b+c+d) - (b+c+d) = 269 - 222 = 47$
- $b = 269 - 208 = 61$
- $c = 269 - 197 = 72$
- $d = 269 - 180 = 89$

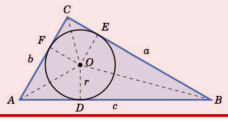
Thus, the largest of the 4 number is 89.

3. A circle inscribed in $\triangle ABC$ has side lengths 8 cm, 10 cm, and 12 cm, as shown in the figure. The length of side BE is equal to:

A) 12 B) 5 C) 68 D) 4

Th: ΔABC inscribed in a circle, let $s = \frac{1}{2}(a,b,c)$ (semi-perimeter)
the radius of the circle:

$$R = \frac{abc}{4\sqrt{s(s-a)(s-b)(s-c)}}$$



$2s = a+b+c$

$2s = (AD+BD)+(BE+CE)+(AF+FC)$

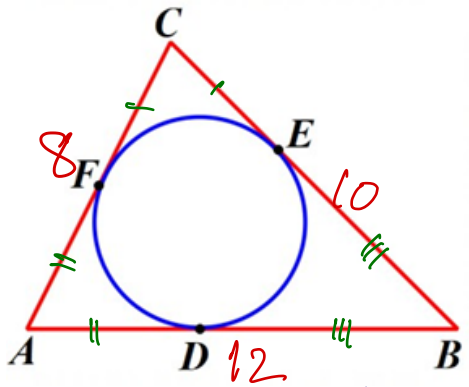
but $AD = AF$, $BE = BD$, $CE = CF$

$\Rightarrow 2s = 2AD + 2BD + 2CE$

$s = AD + \underbrace{BE+CE}_{a}$

$s = AD + a$

$\therefore \boxed{AD = s - a}$



$AD = s - a$

$AD = 15 - 10$

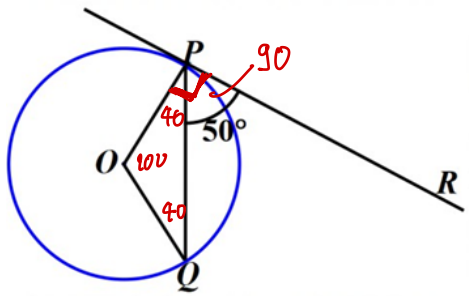
$\boxed{AD = 5}$

$s = \frac{8+10+12}{2} = 15$

$a = 10$

4. In the figure, if O is the center of the circle, PQ is the chord, and the line PR touches the circle at P, forming an angle of 50° with the chord PQ, then $\angle POQ$ is equal to:

A) 100° B) 120° C) 112° D) 130°



5. For a natural number, when it increases by 12, it is equal to 160 times its multiplicative inverse, so that number is:

$\to n$ $\quad n+12 \quad = \quad 160 \times \frac{1}{n}$

A) 26 B) 8 C) 6 D) 32

$n+12 = 160 \times \frac{1}{n}$

$n(n+12) = 160$

$n^2+12n-160=0$

$\Delta = b^2-4ac = 12^2-4(1)(-160) = 784$

$\sqrt{\Delta} = \sqrt{784} = 28$

$n = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-12 \pm 28}{2}$ $\begin{cases} n=-20 \ \times \\ n=8 \ \checkmark \end{cases}$

natural numbers are always positive

or

$n^2-8n+20n-160=0$

$n(n-8)+20(n-8)=0$

$(n-8)(n+20)=0$

$n-8=0,\ n=8 \ \checkmark$

$n+20=0,\ n=-20 \ \times$

6. A takes 10 days less than B's time to complete a task. If A and B can work together to complete the task in 12 days, find the time B alone would take to finish the work.

A) 15 days B) 18 days C) 25 days D) 30 days

$\star$ Work rate $(R) = \frac{\text{Work}}{\text{time}} \Rightarrow \text{Work} = R \times t$

$t_A = t_B - 10$

$R_{A+B} = \frac{1}{12}$ $\to$ 100% of the task $\to$ 12 days to complete

$\Leftrightarrow R_A + R_B = \frac{1}{12}$

$\frac{1}{t_A} + \frac{1}{t_B} = \frac{1}{12}$

$\frac{1}{x} + \frac{1}{x-10} = \frac{1}{12}$

$12(x-10+x) = x(x-10)$

$24x-120 = x^2-10x$

$x^2-34x+120=0$

$(x-30)(x-4)=0$

$x=30,\ x=4 \ \times$

since t_A would be $4-10=-6$

7. At a certain time of sunlight, a 60-meter-tall building casts a 42-meter shadow. How far is the shadow of a 10-meter-high traffic light at the same time?

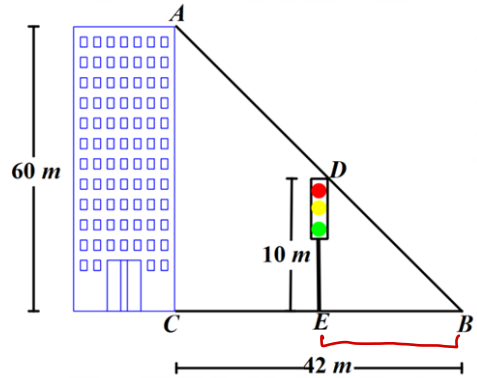
A) 6 m B) 12 m C) 7 m D) 5 m

$\Delta ABC \sim \Delta DBE$

$\frac{AC}{DE} = \frac{BC}{BE}$

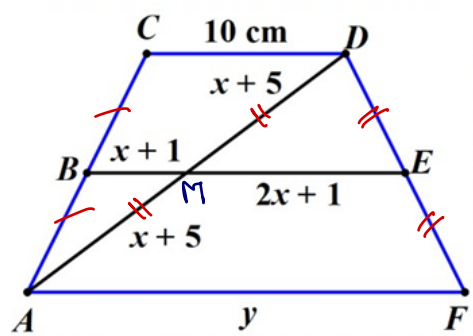
$\frac{60}{10} = \frac{42}{x}$

$x = \frac{\cancel{42}^{7} \times \cancel{10}}{\cancel{60}_{\cancel{10}}} = 7$



8. Determine the length of the diagonal AD in the trapezoid below if $CD \parallel AF$, and B and E are the midpoints of AC and DF, respectively.

A) 16 cm B) 8 cm C) 27 cm D) 18 cm



$\Delta ACD \sim \Delta ABM$

$\frac{CD}{BM} = \frac{AD}{AM}$

$\frac{10}{x+1} = \frac{2\cancel{(x+5)}}{\cancel{x+5}}$

$\frac{10}{x+1} = 2$

$10 = 2x+2$

$5 = x+1$

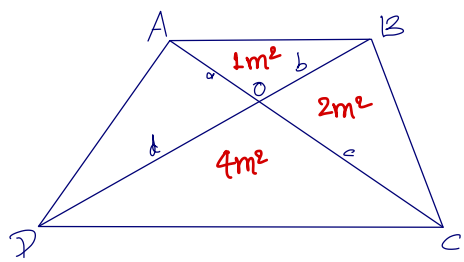
$x = 4$

$\Rightarrow AD = 2(4+5) = 2 \times 9 = \boxed{18}$

quadrilateral

9. Let ~~rectangle~~ $ABCD$ have diagonals AC and BD intersecting at O. If the areas of $\triangle AOB$, $\triangle BOC$, and $\triangle COD$ are equal to $1\,\text{m}^2$, $2\,\text{m}^2$, and $4\,\text{m}^2$ respectively, find the area of $\triangle AOD$ in m^2.

A) $2\,\text{m}^2$ B) $6\,\text{m}^2$ C) $8\,\text{m}^2$ D) $3\,\text{m}^2$



Method 1 : Guess

Draw the quadrilateral,

$A_{\Delta AOD} \simeq 2$ (similar to $\Delta_{\Delta BOC}$ by just looking)

$\Rightarrow$ option $2m^2$ & $3m^2$ are ok

but $3m^2$ would be too large

$\therefore$ $2m^2$ is the correct answer.

Method 2 : Do it the right way!

By proportional of Area of triangle in a quadrilateral :

$\frac{A_{\Delta AOB}}{ab} = \frac{A_{\Delta BOC}}{bc} = \frac{A_{\Delta COD}}{cd} = \frac{A_{\Delta AOD}}{ad} = k$

$\Leftrightarrow \begin{cases} A_{\Delta AOB} = abk = 1 & (1) \\ A_{\Delta BOC} = bck = 2 & (2) \\ A_{\Delta COD} = cdk = 4 & (3) \end{cases}$

where k is constant

$(1) \div (2) : \frac{a}{c} = \frac{1}{2} \Rightarrow c = 2a \; (4)$

$(2) \div (3) : \frac{b}{d} = \frac{1}{2} \Rightarrow d = 2b \; (5)$

From (1) : $abk = 1 \Rightarrow b = \frac{1}{ak}$

sub into (5) : $d = \frac{2}{ak}$

$\therefore A_{\Delta AOD} = kad = k \times a \times \frac{2}{ak} = \boxed{2}$

10. In the trapezoid $ABCD$ below, find the area of the striped part.

A) 25 B) 36 C) 24 D) 16

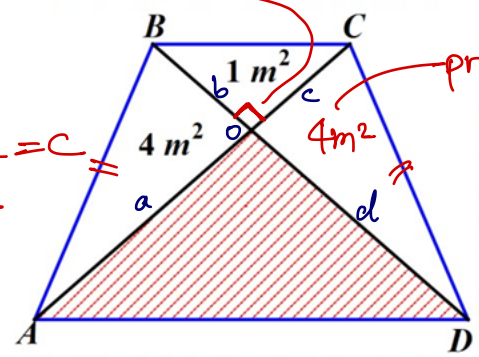
$$\frac{1}{bc} = \frac{4}{ab} = \frac{4}{cd} = \frac{x}{ad} = K$$

$$\begin{cases} \frac{1}{bc} = \frac{4}{ab} \Rightarrow a = 4c \\ \frac{1}{bc} = \frac{4}{cd} \Rightarrow d = 4b \\ \frac{4}{ab} = K \Rightarrow a = \frac{4}{Kb} \end{cases}$$

$$\Rightarrow x = Kad = K\left(\frac{4}{Kb}\right)(4b) = 16$$

$$\therefore \boxed{A_{\Delta AOD} = 16\,m^2}$$

Assuming



probably

$4m^2$

$$\frac{b^2}{2} = 1$$

$$b = \sqrt{2} = c$$

$$\frac{ab}{2} = 4$$

$$ab = 8$$

$$a = \frac{8}{\sqrt{2}}$$

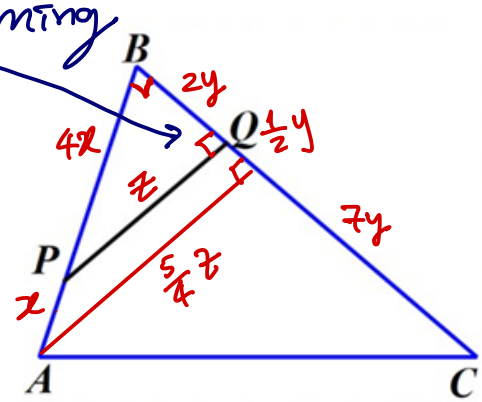
$$a = \frac{8\sqrt{2}}{2} = 4\sqrt{2}$$

$$\Delta AOD = \frac{ad}{2} = \frac{a^2}{2} = \frac{(4\sqrt{2})^2}{2} = \frac{16 \times 2}{2} = \boxed{16}$$

11. From the figure below, $PB = 4AP$, $\frac{BQ}{QC} = \frac{2}{7}$, and the area of the quadrilateral $APQC$ is $37\,\text{cm}^2$. Find the area of triangle BPQ.

A) $10\,\text{cm}^2$ B) $8\,\text{cm}^2$ C) $4\,\text{cm}^2$ D) $12\,\text{cm}^2$

Assuming



$$\frac{\left(\frac{5}{4}z\right)(9y)}{2} - \frac{z(2y)}{2} = 37$$

$$\frac{45yz}{8} - yz = 37$$

$$45yz - 8yz = 37 \times 8$$

$$37yz = 37 \times 8$$

$$yz = \frac{37 \times 8}{37} = 8$$

$$\Delta BPQ = \frac{2yz}{2} = yz = \boxed{8}$$

16. The consecutive points A, B, C and D are on the same straight line where $AB - CD = 22$. If M and N are the midpoints of $[AD]$ and $[BC]$ respectively. Find MN .

A) 16.5 B) 5.5 C) 18 D) 11

Proof. Find MN

- Points A, B, C, and D are on a straight line.
- AB - CD = 22
- M is the midpoint of AD
- N is the midpoint of BC

Imagine a straight line with points A, B, C, and D in that order. M is the midpoint of AD, and N is the midpoint of BC.

Since M is the midpoint of AD, AM = MD. Similarly, BN = NC.

Therefore, we can express AB and CD in terms of these midpoints:

$$AB = AM + MB$$

$$CD = CN + ND$$

Given $AB - CD = 22$

$$(AM + MB) - (CN + ND) = 22$$

Since M and N are midpoints, $AM = MD$ and $BN = NC$. So, we can substitute:

$(MD + MB) - (NC + ND) = 22$ Now, we can rearrange the terms:

$$(MD - ND) + (MB - NC) = 22$$

$MD - ND = (MN + MN)$ (since M and N are midpoints, MN = ND and MN = MB)

$$2MN = 22$$

$$MN = \frac{22}{2} \Rightarrow MN = 11$$

Solution D) 11

17. Simplify expressions $\frac{a^2-9}{5a^3b^3} \div \left[\frac{a+3}{10a^4} \cdot \frac{2a-6}{ab^4}\right]$.

A) ab B) a^2b C) $\frac{ab}{2}$ D) ab^2

Proof. Simplify expressions

We have $\frac{a^2-9}{5a^3b^3} \div \left[\frac{a+3}{10a^4} \cdot \frac{2a-6}{ab^4}\right] = \frac{(a-3)(a+3)}{5a^3b^3} \div \frac{(a+3)(2a-6)}{10a^5b^4}$

$$= \frac{(a-3)(a+3)}{5a^3b^3} \cdot \frac{10a^5b^4}{2(a+3)(a-3)}$$

$$= \frac{10a^5b^4(a-3)(a+3)}{10a^3b^3(a-3)(a+3)} = a^2b$$

Solution B) a^2b

18. If $1 \le k \le 25$. How many prime numbers are there in the form of $6k+1$?

A)18 B) 12 C)16 D) 14

Proof. We know that $1 \le k \le 25$.

Consider If $k=1$ => $6k+1=7$ (it's a prime number)

Consider if $k=25$=> $6k+1=6\times25+1=151$(it's a prime number)

Prime number 7 to151: 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151

K	6k+1	Prime?	K	6k+1	Prime?
1	7	Yes	13	79	Yes
2	13	Yes	14	85	No
3	19	Yes	15	91	No
4	25	No	16	97	Yes
5	31	Yes	17	103	Yes
6	37	Yes	18	109	Yes
7	43	Yes	19	115	No
8	49	No	20	121	No
9	55	No	21	127	Yes
10	61	Yes	22	133	No
11	67	Yes	23	139	Yes
12	73	Yes	24, 25	145, 151	No, Yes

Solution C) 16

19. Ratha reads the same book every day. The table below represents the total number of pages that the Ratha has read at the end of a given number of days.

Number of pages	
Number of days	Total number of pages
2	**32**
4	**64**
5	**80**
7	**112**

How many does Ratha read in one day?

A)16 B) 14 C) 22 D) 15

Proof.

Consider for 2days = 32 pages per day $= \frac{32}{2} = 16$

Just do the same way, Therefore Ratha read 16 pages in a day

Solution A) 16

20. A student has a bus ticket with a total of $30.00. Every time a student takes a bus, the money on the bus ticket is deducted $2.25. How many times can a student ride the bus with the bus ticket?

A)15 B) 14 C)10 D) 13

Proof.

Let x is number of rides

Then $2.25x = 30$

$$x = \frac{30}{2.25} = 13.33$$

Solution D) 13

21. One store sells blue hats and green hats. Each hat costs \$12.00. The expression $12m+12n$ can be used to determine the total prices when a customer purchases the number of blue hats m and the number of green hats n. Which equivalent expression can be used to determine the total price in dollars of the hats?

A) $12mn$ B) $12(m+n)$ C) $6(m+n)$ D) $24mn$

Proof.

The original expression is $12m+12n$

Then we can write it as $12(m+n)$

22. Find the value of $\sqrt{1^3+2^3+3^3+...+10^3}$

A) 79 B) 86 C) 55 D) 66

Proof.

The sum of square formula is $1^2+2^2+3^2+4^2+...+n^2=\frac{n(n+1)(2n+1)}{6}$

The sum of cube formula is $1^3+2^3+3^3+...+n^3=\left(\frac{n(n+1)}{2}\right)^2$

Therefore $\sqrt{1^3+2^3+3^3+...+10^3}=\sqrt{\left(\frac{10(10+1)}{2}\right)^2}=\sqrt{(55)^2}=55$

Solution C) 55

23. Find the value of $(0.000064)^{\frac{2}{3}}\div(0.0016)^{\frac{3}{4}}$

Proof.

We have $\frac{(0.000064)^{\frac{2}{3}}}{(0.0016)^{\frac{3}{4}}}=\frac{(64\times10^{-6})^{\frac{2}{3}}}{(16\times10^{-4})^{\frac{3}{4}}}=\frac{(2^6\times10^{-6})^{\frac{2}{3}}}{(2^4\times10^{-4})^{\frac{3}{4}}}=\frac{2^4\times10^{-4}}{2^3\times10^{-3}}=0.2$

Solution A) 5^{-1}

A) 5^{-1} B) 8^{-1} C) 4^{-1} D) 10^{-1}

24. Ratha is 51 years old and Bopha is 21 years old. How many years will Ratha be twice as old as Bopha?

A) 10 B) 12 C) 6 D) 9

Proof

Let x is the years will Ratha be twice as old as Bopha

Then $(51+x)=2(21+x)$

$51+x=42+2x$

$2x-x=51-42$

$x=9$

Solution A) 9

25. Sophea bought 5 kilograms of oranges, which were 10 oranges per kilogram. She ate eight fruits and gave the rest of her oranges to her six younger siblings. How many oranges did each of her siblings get?

A) 8 B) 7 C) 12 D) 5

Proof.

5kg of oranges $=10\times5=50$ oranges

She ate 8 fruits, Then the remain oranges is $=50-8=42$

The oranges each of her siblings get is $=\frac{42}{6}=7$ oranges

Solution B) 7

26. Find the value of a and b such that $PGCD(a,b)=12$ and $a+b=72$ where $0<a<b$.

A) $a=32, b=12$ B) $a=24, b=60$ C) $a=12, b=60$ D) $a=12, b=24$

Proof.

Since the PGCD of a and b is 12, we can write:

$a=12\times x$

$b=12\times y$

Since $a<b$, Therefore, $x<y$

We know that $a+b=72$. Substituting the expressions for a and b from step 2, we get:

$12x+12y=72$

$x+y=6$

The possible $(x, y) = (1, 5)$ Since x and y is positive integer.

For $(x, y) = (1, 5)$: $a = 12 \times 1 = 12$

$b = 12 \times 5 = 60$

Solution C) a=12, b=60

27. We know the pattern $\frac{1}{1\times 2} = \frac{1}{1} - \frac{1}{2}$, $\frac{1}{2\times 3} = \frac{1}{2} - \frac{1}{3}$, $\frac{1}{3\times 4} = \frac{1}{3} - \frac{1}{4}$. As this pattern, the value of $\frac{1}{1\times 2} + \frac{1}{2\times 3} + \frac{1}{3\times 4} + ... + \frac{1}{2023\times 2024}$ is:

A) 1 B) $\frac{2024}{2023}$ C) $\frac{2023}{2024}$ D) $\frac{1}{2}$

Proof.

$$\frac{1}{1\times 2} + \frac{1}{2\times 3} + \frac{1}{3\times 4} + ... + \frac{1}{2023\times 2024} = \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + ... - \frac{1}{2023} + \frac{1}{2023} - \frac{1}{2024}$$

$$= 1 - \frac{1}{2024} = \frac{2023}{2024}$$

Solution C) $\frac{2023}{2024}$

28. Sok and Sao invested \$12,100 in business. $\frac{3}{5}$ of Sok's share is equal to $\frac{7}{9}$ of Sao's share. How much is Sao's shareholding?

A) \$5,269 B) \$6,831 C) \$5,423 D) \$5,420

Proof.

Let's denote Sok's share as S and Sao's share as Sa.

We have: $S + Sa = 12100$ (1)

And $\frac{3}{5}S = \frac{7}{9}Sa$ (2)

From the second equation, we can express S in terms of Sa:

Therefore $S = \frac{35}{27}Sa$

Eq (1) $\frac{35}{27}Sa + Sa = 12100$

$\frac{62}{27}Sa = 12100$

$Sa = \frac{12100 \times 27}{62} = \$5269.35$

Thus $Sa \approx \$5269$

Solution A) $5,269

29. Last week, the price of apple at a grocery store was $1.60 per kilogram. This week, apples at the same grocery store are on sale for 10% off. What is the total price of 4kg of apples this week at the grocery store?

A) $6.93 B) $6.75 C) $4.77 D) $6.48

Proof.

Original price per kg= $\$1.60$

Price after discount 10% is $= \$1.60(1-0.1) = \$1.44$

Therefore, the total price of 4kg is $= 4 \times 1.44 = \$5.76$

Solution $5.76

30. A worker arrives at work 15 minutes late as he walks at 4 kilometers per hour from home. The next day, he accelerates 2 km/h and arrives at work on time. Find the distance from his home to his workplace.

A) 5km B) 3km C) 4km D) 2km

Proof. Let's denote the distance from home to workplace as D.

- First day: The worker took 15 mn longer to travel with the speed of 4km/h

Therefore, $D = 4(t + 0.25)$, where is the time taken to travel without being late.

- Second day: The worker traveled at speed of 6 km/h on time

Then $D = 6t$

Equating both expressions for D: $4(t + 0.25) = 6t$

$4t + 1 = 6t \Rightarrow t = \frac{1}{2} = 0.5$ Hour

Therefore, the distance from the worker's home to their workplace is $D = 6 \times 0.5 = 3km$

Solution B) 3km